

WASP-39b

R-1673

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_150521 · created 2026-08-02T10:46:18+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457163.794 +/- 0.045 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.055
Transit depth (Rp/Rs)^2	3.6 +/- 2.08 [%]
Orbital Inclination (inc)	89.07 +/- 2.09
Airmass coefficient 1 (a1)	0.76 +/- 0.45
Airmass coefficient 2 (a2)	0.06 +/- 0.4
Scatter in the residuals of the lightcurve fit is	59.31 %
Best Comparison Star	None
Optimal Aperture	2.05
Optimal Annulus	6.5
Transit Duration (day)	0.114 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.19 $\pm$ 0.055 vs 0.1457 (deviation 0.81 σ)
Duration fit vs published	0.114 d vs 0.12 d (deviation 0.24 σ)
Mid-time O–C	5.2 min
Depth SNR	3.5
Residual scatter	59.31 %

Light curve

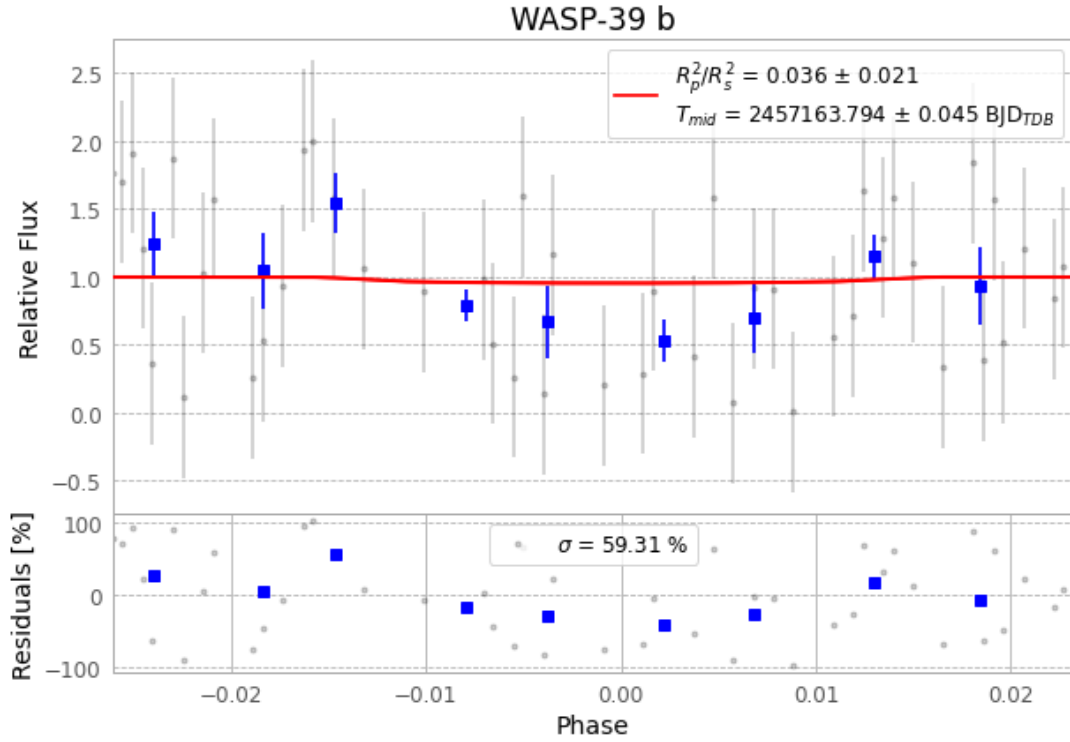


Figure 1 — FinalLightCurve_WASP-39 b_2015-05-20.png (SHA-256 ccc9afefe4f6c2cb...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [368, 167], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b1c557986531f169...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

97 FITS frames (64 MB), WASP-39150521042112.FITS → WASP-39150521090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-150521.log (bda3361ca1bf...), FinalLightCurve_WASP-39 b_2015-05-20.png (ccc9afefe4f6...), FinalLightCurve_WASP-39 b_2015-05-20.csv (1bad0fb4b876...)

Compute resources

Wall time 155.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:46Z.